Alexandre Garcia-Duran

agarcia@crm.cat | +34 630 663 306 | LinkedIn | ORCID: 0009-0004-2473-8111

RESEARCH SUMMARY

PhD researcher in **computational neuroscience** at the Centre de Recerca Matemàtica (CRM, Barcelona). My research focuses on understanding **vision**: I use **probabilistic graphical models, machine learning, and dynamical systems theory** to study how the brain performs perceptual inference, particularly through the lens of bistable perception. I am also interested in **adversarial robustness** and **connectome-constrained** neural network models, which I explored during a recent visit to the Turaga lab at Janelia Research Campus (HHMI).

EDUCATION

PhD Applied Mathematics *2023 – Present*

Centre de Recerca Matemàtica (CRM) and Universitat Politècnica de Catalunya (UPC). Computational Neuroscience Group. Supervised by A. Hyafil & M. Molano-Mazón.

MSc Advanced Mathematics and Mathematical Engineering *2022 – 2023*

Universitat Politècnica de Catalunya (UPC).

- Thesis: "An integrative model for perceptual decision-making and movement". Grade 10/10 (Honors).
- GPA 9.1 / 10.

BEng Biomedical Engineering *2018 – 2022*

Universitat de Barcelona (UB).

- Thesis: "Disentangling human decisions under strong time pressure in an expectation-based 2AFC auditory task". Grade 9.7/10.
- Honors in 4 subjects: Internship program, Biomedical Signals & Instrumentation, Biological Systems Modelling, and Biomechanics.
- GPA 8.2 / 10.

RESEARCH EXPERIENCE

Doctoral Researcher · Centre de Recerca Matemàtica (CRM) *Nov 2023 – Present*

Computational Neuroscience Group. Supervised by A. Hyafil & M. Molano-Mazón.

- Developing **probabilistic graphical models** of visual perception, using **Bayesian inference**.
- Studying how **lateral interactions** in the visual hierarchy encode statistical structure for perceptual inference.
- Presented at major conferences: COSYNE, CCN, ICMNS (5 posters + 1 invited talk in 2024–2026).

Short-term Research Visitor · HHMI Janelia Research Campus *Mar – Apr 2026*

Turaga Lab.

- Investigated **adversarial attacks** on optical-flow networks constrained by the Drosophila **connectome**.
- Explored computational links between fly visual illusions and network robustness.
- Host: Srinivas C. Turaga.

Research Assistant · Centre de Recerca Matemàtica (CRM)
Jul 2022 – Oct 2023

Computational Neuroscience Group. Supervised by A. Hyafil, M. Molano-Mazón & G. Huguet.

- Led behavioral analysis and modelling for rat and human sensorimotor data; used **drift-diffusion models**.
- Co-authored a paper published in **Nature Communications** (2024).

Research Intern · IDIBAPS
Jan – Jun 2022

Brain Circuits & Behavior Lab. Group of Jaime de la Rocha.

- Conducted psychophysics experiments and analyzed human behavioral data (history biases, GLMs).

Research Intern · Institute for Bioengineering of Catalonia (IBEC)
Jun – Aug 2021

Integrative Cell & Tissue Dynamics lab. Supervised by Nimesh Chahare and Xavier Trepat.

- Worked on mechanobiology and microfluidics experiments.

PUBLICATIONS

Garcia-Duran A., Wokke M., Pou-Amengual N., Molano-Mazón M., Hyafil A. “Bistable perception reveals a canonical computation in sensory processing”. 2026. *In preparation*.

Garcia-Duran* A., Rodriguez-Garcia* A., Molano-Mazon M., Hyafil A., Ramaswamy S., Lappalainen J. K. “Measuring Robustness and Efficiency in a Connectome-Constrained Fly Visual System Model on a Collision-Detection Task”. *Under review. 2026. (* equal contribution)*

Molano-Mazón*, M., **Garcia-Duran*, A.**, Pastor-Ciurana*, J., Hernández-Navarro, Ll., Bektic, L., Lombardo, D., De la Rocha, J., Hyafil, A. “Rapid, systematic updating of movement by accumulated decision evidence”. *Nature communications*, 15, 10583; 2024. DOI *(* equal contribution)*

SELECTED POSTERS & TALKS

Invited talks

- "Bistable perception reveals a canonical computation in sensory processing." BARCCSYN, Barcelona, May 2026.
- "Bistable perception reveals a canonical computation in sensory processing." Neuroscience Seminars at UPC, May 2026.
- "An attempt to understand (bistable) perception." SIJIMAT seminar, CRM, Barcelona, June 2025.
- "Bistable perception emerges from loopy inference in strongly coupled probabilistic graphs." Neuroscience Seminars at UPC, December 2024.

Selected Posters (total: 10)

- COSYNE 2025 (Lisbon, Mar 2025). "Lateral connections in visual hierarchy encode statistical structure to support perceptual inference."
- CCN 2025 (Amsterdam, Aug 2025). "Bistable perception emerges from loopy inference in strongly coupled probabilistic graphs."
- COSYNE 2024 (Lisbon, Feb 2024). "Rapid, systematic updating of movement by accumulated decision evidence."
- IBRO World Congress of Neuroscience 2023 (Granada, Sep 2023). "Fast and tight control of motor vigor by accumulated decision evidence."

FELLOWSHIPS & GRANTS

PhD Fellowship · Spanish Ministry of Science and Innovation — *Nov 2023 – Nov 2027*

"Deciphering the brain mechanisms for perceptual inference through bistable perception."

Computational Resource Grant, National Science Foundation (NSF) ACCESS Program

Project ID: CIS261112

"FlyAdVis: Adversarial robustness and motion detection disruptions in a connectome-constrained fly visual system model."

Duration: 2026 – 2027

Allocation: 200K ACCESS Credits (~3000 GPU hours)

TECHNICAL SKILLS

• Python (primary — NumPy, PyTorch, scikit-learn) • MATLAB · R • Bayesian networks / Probabilistic ML • Neural Networks (CNN, ANN, LLM) • Adversarial attacks & robustness analysis	• Connectome-constrained neural networks • Dynamical systems • Psychophysics experimental design • Optical flow & image processing • Variational autoencoders & Boltzmann Machines

STUDENT SUPERVISION

Supervised two master students.

TRAINING & CERTIFICATES

- CAJAL Advanced Neuroscience Training Programme – NeuroAI – Neuroscience and AI, Lisbon, Portugal. July 2025.
- Barcelona Summer School for Advanced Modelling of Behaviour (BAMB!) 2024.
- Probabilistic Graphical Models Specialization: Coursera / Stanford Online (2024).
- Deep Learning Specialization: Coursera / DeepLearning.AI (2021).

COMMUNITY & SERVICE

- Co-organizer of Neurochats seminars (CRM, Barcelona): an interdisciplinary forum for PhD students and postdocs.
- Co-organizer of BARCCSYN 2025 annual meeting (Barcelona, May 2025).
- BIYSC teacher. Short lessons of graph theory. July 2024.

Last updated: July 2026